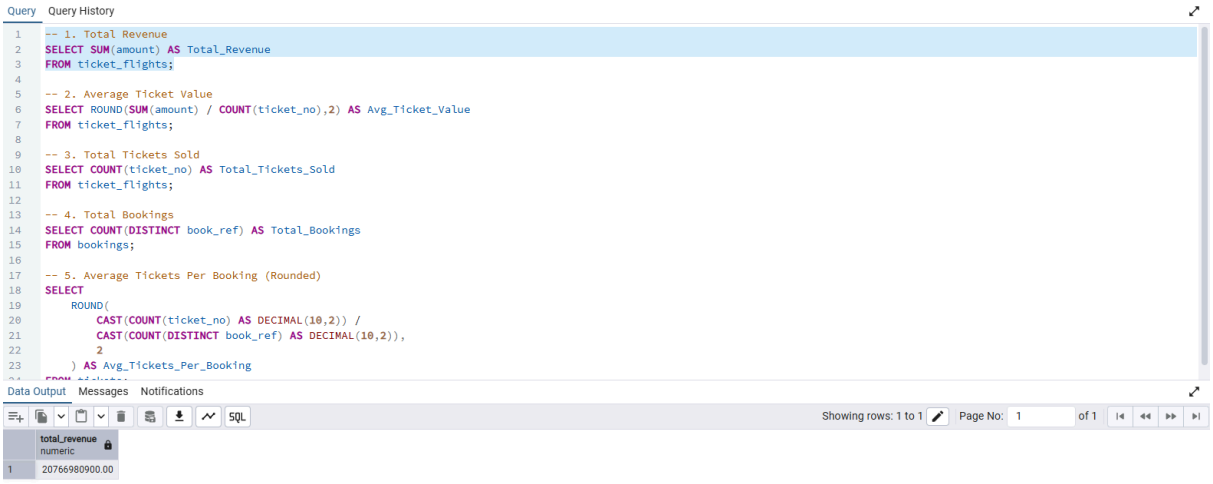
# AIRLINE DATA EXTRACT & VALIDATION – POSTGRESQL

## 1. Total Revenue

*Objective: Calculate the total monetary value of all booked flights.*

```
SELECT SUM(amount) AS Total_Revenue
FROM ticket_flights;
```




## 2. Average Ticket Value

*Objective: Determine the average revenue generated per individual ticket.*

```
SELECT CAST(SUM(amount) / COUNT(ticket_no) AS DECIMAL(10,
2)) AS Avg_Ticket_Value
FROM ticket_flights;
```

Query Query History

```sql
-- 1. Total Revenue
SELECT SUM(amount) AS Total_Revenue
FROM ticket_flights;

-- 2. Average Ticket Value
SELECT ROUND(SUM(amount) / COUNT(ticket_no),2) AS Avg_Ticket_Value
FROM ticket_flights;

-- 3. Total Tickets Sold
SELECT COUNT(ticket_no) AS Total_Tickets_Sold
FROM ticket_flights;

-- 4. Total Bookings
SELECT COUNT(DISTINCT book_ref) AS Total_Bookings
FROM bookings;

-- 5. Average Tickets Per Booking (Rounded)
SELECT
    ROUND(
        CAST(COUNT(ticket_no) AS DECIMAL(10,2)) /
        CAST(COUNT(DISTINCT book_ref) AS DECIMAL(10,2)),
        2
    ) AS Avg_Tickets_Per_Booking
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

| | avg_ticket_value numeric |
|---|---|
| 1 | 19858.91 |

## 3. Total Bookings & Total Tickets Sold

*Objective: Quantify total sales volume and booking transactions.*

```sql
SELECT
(SELECT COUNT(ticket_no) FROM ticket_flights) AS Total_Tickets_Sold,
(SELECT COUNT(DISTINCT book_ref) FROM bookings) AS Total_Bookings;
```

Query Query History

```sql
-- 1. Total Revenue
SELECT SUM(amount) AS Total_Revenue
FROM ticket_flights;

-- 2. Average Ticket Value
SELECT ROUND(SUM(amount) / COUNT(ticket_no),2) AS Avg_Ticket_Value
FROM ticket_flights;

-- 3. Total Tickets Sold
SELECT COUNT(ticket_no) AS Total_Tickets_Sold
FROM ticket_flights;

-- 4. Total Bookings
SELECT COUNT(DISTINCT book_ref) AS Total_Bookings
FROM bookings;

-- 5. Average Tickets Per Booking (Rounded)
SELECT
    ROUND(
        CAST(COUNT(ticket_no) AS DECIMAL(10,2)) /
        CAST(COUNT(DISTINCT book_ref) AS DECIMAL(10,2)),
        2
    ) AS Avg_Tickets_Per_Booking
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

| | total_tickets_sold bigint |
|---|---|
| 1 | 1045726 |

Query Query History

```sql
-- 1. Total Revenue
SELECT SUM(amount) AS Total_Revenue
FROM ticket_flights;

-- 2. Average Ticket Value
SELECT ROUND(SUM(amount) / COUNT(ticket_no),2) AS Avg_Ticket_Value
FROM ticket_flights;

-- 3. Total Tickets Sold
SELECT COUNT(ticket_no) AS Total_Tickets_Sold
FROM ticket_flights;

-- 4. Total Bookings
SELECT COUNT(DISTINCT book_ref) AS Total_Bookings
FROM bookings;

-- 5. Average Tickets Per Booking (Rounded)
SELECT
    ROUND(
        CAST(COUNT(ticket_no) AS DECIMAL(10,2)) /
        CAST(COUNT(DISTINCT book_ref) AS DECIMAL(10,2)),
        2
    ) AS Avg_Tickets_Per_Booking
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

| | total_bookings bigint |
|---|---|
| 1 | 262788 |

## 3. Average Tickets Per Booking

*Objective: Determine the average tickets booked per booking.*

```sql
SELECT
ROUND (
CAST AS DECIMAL (10, 2)) /
CAST (COUNT(DISTINCT book_ref) AS DECIMAL (10,2)) ,
2
) AS Avg_Tickets_Per_Booking
FROM tickets;
```

Query Query History

```sql
SELECT SUM(amount) AS Total_Revenue
FROM ticket_flights;

-- 2. Average Ticket Value
SELECT ROUND(SUM(amount) / COUNT(ticket_no),2) AS Avg_Ticket_Value
FROM ticket_flights;

-- 3. Total Tickets Sold
SELECT COUNT(ticket_no) AS Total_Tickets_Sold
FROM ticket_flights;

-- 4. Total Bookings
SELECT COUNT(DISTINCT book_ref) AS Total_Bookings
FROM bookings;

-- 5. Average Tickets Per Booking (Rounded)
SELECT
    ROUND(
        CAST(COUNT(ticket_no) AS DECIMAL(10,2)) /
        CAST(COUNT(DISTINCT book_ref) AS DECIMAL(10,2)),
        2
    ) AS Avg_Tickets_Per_Booking
FROM tickets;
```

Data Output Messages Notifications

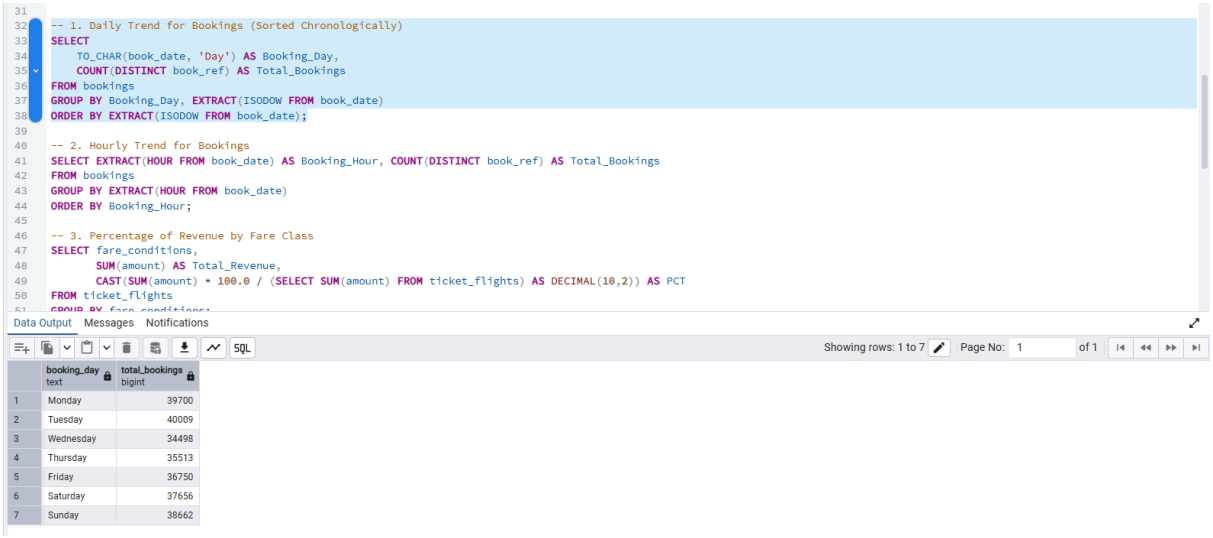
Showing rows: 1 to 1 Page No: 1 of 1

| | avg_tickets_per_booking numeric |
|---|---|
| 1 | 1.40 |

## 4. Booking Volume Trends (Daily & Hourly)

*Objective: Identify peak booking periods for marketing and server load optimization.*

```
- Daily Trend
SELECT TO_CHAR(book_date, 'Day') AS Booking_Day, COUNT(DIST
INCT book_ref) AS Total_Bookings
FROM bookings
GROUP BY TO_CHAR(book_date, 'Day')
ORDER BY Total_Bookings DESC;
```



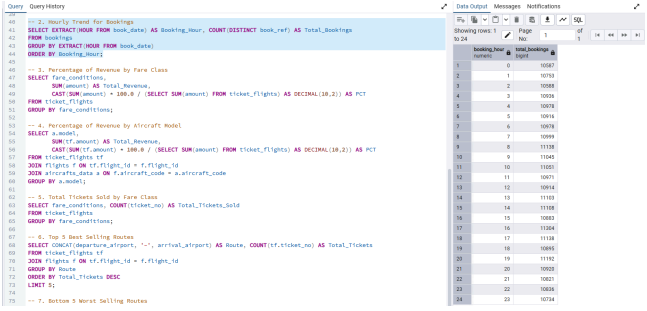
```
-- 1. Daily Trend for Bookings (Sorted Chronologically)
SELECT
    TO_CHAR(book_date, 'Day') AS Booking_Day,
    COUNT(DISTINCT book_ref) AS Total_Bookings
FROM bookings
GROUP BY Booking_Day, EXTRACT(ISODOW FROM book_date)
ORDER BY EXTRACT(ISODOW FROM book_date);

-- 2. Hourly Trend for Bookings
SELECT EXTRACT(HOUR FROM book_date) AS Booking_Hour, COUNT(DISTINCT book_ref) AS Total_Bookings
FROM bookings
GROUP BY EXTRACT(HOUR FROM book_date)
ORDER BY Booking_Hour;

-- 3. Percentage of Revenue by Fare Class
SELECT fare_conditions,
       SUM(amount) AS Total_Revenue,
       CAST(SUM(amount) * 100.0 / (SELECT SUM(amount) FROM ticket_flights) AS DECIMAL(10,2)) AS PCT
FROM ticket_flights
```

Data Output Messages Notifications

Showing rows: 1 to 7 Page No: 1 of 1

| | booking_day text | total_bookings bigint |
|---|---|---|
| 1 | Monday | 39700 |
| 2 | Tuesday | 40009 |
| 3 | Wednesday | 34498 |
| 4 | Thursday | 35513 |
| 5 | Friday | 36750 |
| 6 | Saturday | 37656 |
| 7 | Sunday | 38662 |

```
- Hourly Trend
SELECT EXTRACT(HOUR FROM book_date) AS Booking_Hour, COUNT
(DISTINCT book_ref) AS Total_Bookings
FROM bookings
GROUP BY EXTRACT(HOUR FROM book_date)
ORDER BY Booking_Hour;
```



## 5. Revenue Distribution by Fare Class

*Objective: Analyze which cabin classes drive the most revenue.*

```
SELECT fare_conditions,
SUM(amount) AS Total_Revenue,
CAST(SUM(amount) * 100.0 / (SELECT SUM(amount) FROM ticket_
flights) AS DECIMAL(10,2)) AS Revenue_Percentage
FROM ticket_flights
GROUP BY fare_conditions;
```

```
-- 5. Total Tickets Sold by Fare Class
SELECT fare_conditions, COUNT(ticket_no) AS Total_Tickets_Sold
FROM ticket_flights
GROUP BY fare_conditions;

-- 6. Top 5 Best Selling Routes
SELECT CONCAT(departure_airport, '-', arrival_airport) AS Route, COUNT(tf.ticket_no) AS Total_Tickets
FROM ticket_flights tf
JOIN flights f ON tf.flight_id = f.flight_id
GROUP BY Route
ORDER BY Total_Tickets DESC
LIMIT 5;

-- 7. Bottom 5 Worst Selling Routes
SELECT CONCAT(departure_airport, '-', arrival_airport) AS Route, COUNT(tf.ticket_no) AS Total_Tickets
FROM ticket_flights tf
JOIN flights f ON tf.flight_id = f.flight_id
GROUP BY Route
ORDER BY Total_Tickets ASC
LIMIT 5;

--CHART QUERIES ENDS HERE
```

Data Output | Messages | Notifications

Showing rows: 1 to 3 Page No: 1 of 1

| | fare_conditions character varying (10) | total_tickets_sold bigint |
|---|---|---|
| 1 | Business | 107642 |
| 2 | Comfort | 17291 |
| 3 | Economy | 920793 |

## 6. % Revenue Distribution by Fare Class

*Objective: Analyze which cabin classes drive the most revenue by percentage.*

```
SELECT a.model,
SUM(tf. amount) AS Total_Revenue,
CAST (SUM(tf. amount) * 100.0 / (SELECT SUM(amount) FROM ti
cket_flights) AS DECIMAL (10,2)) AS PCT
FROM ticket_flights tf
JOIN flights f ON tf.flght_id
f.flight_id
JOIN aircrafts_data a ON f. aircraft_code = a.aircraft_code
GROUP BY a.model;
```

```sql
-- 3. Percentage of Revenue by Fare Class
SELECT fare_conditions,
       SUM(amount) AS Total_Revenue,
       CAST(SUM(amount) * 100.0 / (SELECT SUM(amount) FROM ticket_flights) AS DECIMAL(10,2)) AS PCT
FROM ticket_flights
GROUP BY fare_conditions;

-- 4. Percentage of Revenue by Aircraft Model
SELECT a.model,
       SUM(tf.amount) AS Total_Revenue,
       CAST(SUM(tf.amount) * 100.0 / (SELECT SUM(amount) FROM ticket_flights) AS DECIMAL(10,2)) AS PCT
FROM ticket_flights tf
JOIN flights f ON tf.flight_id = f.flight_id
JOIN aircrafts_data a ON f.aircraft_code = a.aircraft_code
GROUP BY a.model;

-- 5. Total Tickets Sold by Fare Class
SELECT fare_conditions, COUNT(ticket_no) AS Total_Tickets_Sold
FROM ticket_flights
GROUP BY fare_conditions;
```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

| | fare_conditions character varying (10) | total_revenue numeric | pct numeric (10,2) |
|---|---|---|---|
| 1 | Business | 5505179600.00 | 26.51 |
| 2 | Comfort | 566116900.00 | 2.73 |
| 3 | Economy | 14695684400.00 | 70.76 |

## 7. Revenue Distribution by Aircraft Model

*Objective: Analyze which aircraft drives the most revenue.*

```sql
SELECT a.model,
SUM(tf. amount) AS Total _ Revenue,
CAST(tf.amount) * 100.0 / (SELECT SUM(amount) FROM ticket_f
tights) AS DECIMAL (10,2)) AS PCT
FROM ticket _ flights tf
JOIN flights f ON tf. flight_id
f. flight_id
JOIN aircrafts_data a ON f.aircraft_code
= a.aircraft code
GROUP BY a.model;
```

Query Query History

```sql
-- 4. Percentage of Revenue by Aircraft Model
SELECT a.model,
       SUM(tf.amount) AS Total_Revenue,
       CAST(SUM(tf.amount) * 100.0 / (SELECT SUM(amount) FROM ticket_flights) AS DECIMAL(10,2)) AS PCT
FROM ticket_flights tf
JOIN flights f ON tf.flight_id = f.flight_id
JOIN aircrafts_data a ON f.aircraft_code = a.aircraft_code
GROUP BY a.model;

-- 5. Total Tickets Sold by Fare Class
SELECT fare_conditions, COUNT(ticket_no) AS Total_Tickets_Sold
FROM ticket_flights
GROUP BY fare_conditions;

-- 6. Top 5 Best Selling Routes
SELECT CONCAT(departure_airport, '-', arrival_airport) AS Route, COUNT(tf.ticket_no) AS Total_Tickets
FROM ticket_flights tf
JOIN flights f ON tf.flight_id = f.flight_id
GROUP BY Route
ORDER BY Total_Tickets DESC
LIMIT 5;

-- 7. Bottom 5 Worst Selling Routes
SELECT CONCAT(departure_airport, '-', arrival_airport) AS Route, COUNT(tf.ticket_no) AS Total_Tickets
```

Data Output Messages Notifications

Showing rows: 1 to 8 Page No: 1 of 1

| | model character varying (50) | total_revenue numeric | pct numeric (10,2) |
|---|---|---|---|
| 1 | Airbus A319-100 | 2706163100.00 | 13.03 |
| 2 | Airbus A321-200 | 1638164100.00 | 7.89 |
| 3 | Boeing 737-300 | 1426552100.00 | 6.87 |
| 4 | Boeing 767-300 | 4371277100.00 | 21.05 |
| 5 | Boeing 777-300 | 3431205500.00 | 16.52 |
| 6 | Bombardier CRJ-200 | 1982760500.00 | 9.55 |
| 7 | Cessna 208 Caravan | 96373800.00 | 0.46 |
| 8 | Sukhoi Superjet-100 | 5114484700.00 | 24.63 |

## 8. Route Performance (Top & Bottom 5)

*Objective: Highlight the most and least successful flight paths.*

```sql
- Top 5 Routes
SELECT CONCAT(departure_airport, '-', arrival_airport) AS F
light_Route, COUNT(tf.ticket_no) AS Total_Tickets
FROM ticket_flights tf
JOIN flights f ON tf.flight_id = f.flight_id
GROUP BY Flight_Route
ORDER BY Total_Tickets DESC
LIMIT 5;
```

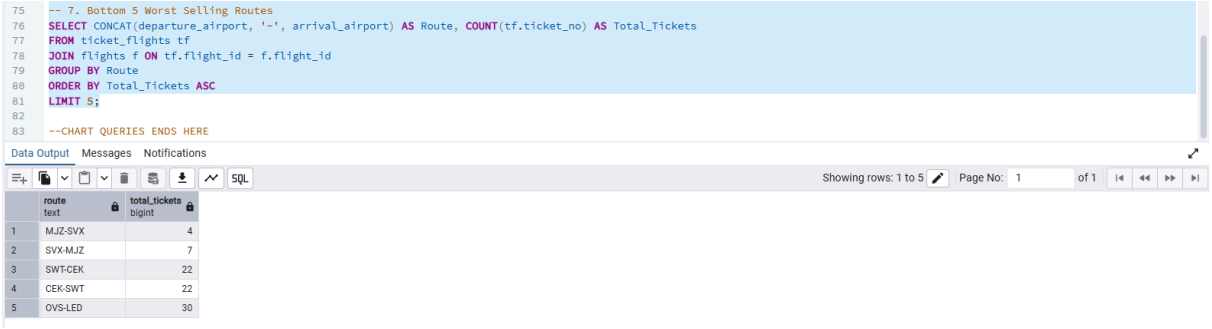
```sql
-- 6. Top 5 Best Selling Routes
SELECT CONCAT(departure_airport, '-', arrival_airport) AS Route, COUNT(tf.ticket_no) AS Total_Tickets
FROM ticket_flights tf
JOIN flights f ON tf.flight_id = f.flight_id
GROUP BY Route
ORDER BY Total_Tickets DESC
LIMIT 5;

-- 7. Bottom 5 Worst Selling Routes
SELECT CONCAT(departure_airport, '-', arrival_airport) AS Route, COUNT(tf.ticket_no) AS Total_Tickets
FROM ticket_flights tf
JOIN flights f ON tf.flight_id = f.flight_id
GROUP BY Route
ORDER BY Total_Tickets ASC
LIMIT 5;

--CHART QUERIES ENDS HERE
```

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

| | route text | total_tickets bigint |
|---|---|---|
| 1 | SVO-LED | 16461 |
| 2 | DME-OVB | 15903 |
| 3 | SVO-SVX | 15879 |
| 4 | SVX-SVO | 15433 |
| 5 | LED-SVO | 15424 |

```
- Bottom 5 Routes
SELECT CONCAT(departure_airport, '-', arrival_airport) AS F
light_Route, COUNT(tf.ticket_no) AS Total_Tickets
FROM ticket_flights tf
JOIN flights f ON tf.flight_id = f.flight_id
GROUP BY Flight_Route
ORDER BY Total_Tickets ASC
LIMIT 5;
```



```
-- 7. Bottom 5 Worst Selling Routes
SELECT CONCAT(departure_airport, '-', arrival_airport) AS Route, COUNT(tf.ticket_no) AS Total_Tickets
FROM ticket_flights tf
JOIN flights f ON tf.flight_id = f.flight_id
GROUP BY Route
ORDER BY Total_Tickets ASC
LIMIT 5;

--CHART QUERIES ENDS HERE
```

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

| | route text | total_tickets bigint |
|---|---|---|
| 1 | MJZ-SVX | 4 |
| 2 | SVX-MJZ | 7 |
| 3 | SWT-CEK | 22 |
| 4 | CEK-SWT | 22 |
| 5 | OVS-LED | 30 |